Matthew YC Cheung

Ph.D. Candidate, Electrical and Computer Engineering

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Summary

Machine learning researcher and Rice University ECE Ph.D. candidate (expected Dec. 2026) developing methods to evaluate and improve AI systems. First-author research at EMNLP (Outstanding Paper; Oral), ICLR, and MICCAI (Spotlight) spanning LLM reasoning, statistically rigorous model evaluation, computer vision, and generative imaging. Industry experience building medical-imaging ML models for Phase 3 clinical-trial applications and software for manufacturing R&D.

Education

Ph.D. **Rice University**, Electrical and Computer Engineering — Houston, TX — Aug 2023 – Dec 2026

- Advisors: Ashok Veeraraghavan (Rice Computational Imaging Lab) and Guha Balakrishnan (Rice Vision and Imaging Group)
- Trainee in the NIH NLM Training Program in Biomedical Informatics and Data Science
- Thesis: Exploiting Structure in Learning Pipelines for Practical Conformal Inference
- Topics of interest: computer vision, conformal prediction, language modeling, reasoning verification

M.S. **Rice University**, Electrical and Computer Engineering — Houston, TX — Aug 2020 – Dec 2023

- Advisor: Ashok Veeraraghavan (Rice Computational Imaging Lab)
- Thesis: Wearable Blood Pressure Monitoring Devices: Understanding Heterogeneity in Design and Evaluation
- Teaching assistant for undergraduate and graduate courses in computer vision and linear algebra

B.S. **University of California, Davis**, Electrical Engineering — Davis, CA — Sept 2015 – June 2020

- Graduated cum laude
- Graduate coursework in optical microscopy, large-scale scientific computing, computational imaging, and solid-state device physics
- Tau Beta Pi, CA Lambda Chapter

Experience

Johnson & Johnson R&D, AI/ML Intern, Computer Vision — Remote — May 2026 – Aug 2026 — 4 months

Developed deep-learning models for ophthalmic imaging, now being deployed in Phase 3 clinical trials.

- Designed morphology-aware losses for geographic-atrophy biomarker quantification from volumetric OCT (GAIZ, MLMI @ MICCAI 2026 Oral)
- Trained segmentation models and developed a data-augmentation approach for fundus autofluorescence imaging

Rice University, Graduate Research Assistant — Houston, TX — Dec 2020 – present — 6 years

Advised by Ashok Veeraraghavan and Guha Balakrishnan.

- LLM reasoning reliability: verifier-agnostic calibration for certifying reasoning-trace prefixes (CROP, EMNLP 2026 Oral, Outstanding Paper)
- Vision reliability: statistically valid uncertainty intervals on segmentation metrics (COMPASS, ICLR 2026; ConVOLT, MICCAI 2026 Spotlight)
- Developed uncertainty quantification for probabilistic image reconstruction (Metric-Guided, UNSURE @ MICCAI 2025 Oral)
- Mentored 6 undergraduate researchers and supervised 2 data-science capstone projects

Texas Instruments, Quality Reliability Engineering Intern — Dallas, TX
- Applied statistical drift analysis to high-voltage gate-driver reliability and developed system-level testing procedures
- Built data-cleaning, clustering, and database automation tools for production quality workflows

June 2019 – Sept 2019
4 months

Yang Research Laboratory, Undergraduate Research Assistant — Davis, CA
- Simulated optical setups for classification and differentiation with spatial light modulators in TensorFlow

Sept 2018 – July 2020
1 year 11 months

Enovix Corporation, Software Development Intern (R&D, Engineering) — Fremont, CA
- Built Python tools for battery characterization, metrology, and manufacturing process control
- Automated production-data extraction and visualization, and maintained SQL databases

June 2017 – Jan 2019
1 year 8 months

Kovnir Group, Undergraduate Research Assistant — Davis, CA
- Synthesized type-I clathrate compounds and characterized samples with powder X-ray diffraction

Mar 2016 – May 2017
1 year 3 months

Skills

Programming and ML frameworks: Python, PyTorch, Hugging Face Transformers/Diffusers, SciPy, NumPy, Pandas, scikit-learn

Generative and vision models: Transformers, language models, vision-language models, diffusion models, segmentation, reconstruction, object detection, registration

Statistical ML: Conformal inference, risk control, uncertainty quantification, calibration, distribution shift, model evaluation

Language model evaluation: Reasoning verification, process supervision, model benchmarking, selective prediction

Mentoring

- Adilkhan Kairzhan (Rice ECE, May 2025 – Present)
- Leslie Espinoza (Rice Statistics, Apr. 2025 – Aug. 2025)
- Luke Yuen (Rice MECH, REU, May 2023 – Dec. 2024; now pursuing an M.S. in Mechanical Engineering at Stanford)
- Rachel Gu (Rice ECE, REU, May 2023 – Dec. 2024; ECE Distinction in Research and Creative Work; now Engineer at Tetra Tech)
- Cindy Pineda (Lone Star Community College Biotechnology, NSF START, Nov. 2023 – Aug. 2024)
- Desiree Gutierrez (Texas A&M, PATHS-UP REU, May 2023 – Jul. 2023)

Awards

D2K Mentoring Fellowship — Aug 2024
Rice University

NLM Training Program in Biomedical Informatics and Data Science — June 2023
National Library of Medicine

PATHS-UP Fellow — Jan 2022
PATHS-UP

PATHS-UP Student Seed Fund Award — Oct 2023
PATHS-UP

Graduate Student Distinguished Mentoring Fellowship — Oct 2023
Rice University

Rice Fellow Aug 2020
Rice University

John Eric Peckham Memorial Scholarship Sept 2019
UC Davis Undergraduate and Prestigious Scholarships

Joseph Bonnheim Memorial Scholarship Sept 2016
UC Davis Undergraduate and Prestigious Scholarships

Outstanding Senior Design Project Award May 2019
UC Davis Department of Electrical & Computer Engineering

Texas Instruments Scholar Award May 2019
UC Davis Department of Electrical & Computer Engineering

Arthur & Julia Suran Endowment Scholarship Aug 2018
UC Davis College of Engineering

Scholarship for Outstanding Performance in Major May 2016
UC Davis Department of Chemistry

Kraft Prize Jan 2016
UC Davis Undergraduate Scholarships, Honors and Prizes

ESF Chairman's Award for Academic Excellence Aug 2015
English Schools Foundation